

Cognitive Frameworks in Decision-Making and Problem-Solving: A Critical Reflective Analysis

Introduction

Human decision-making and problem-solving are complex cognitive processes influenced by internal psychological frameworks and external environmental constraints (Halpern & Dunn, 2023). While classical normative models often prescribe optimal rationality, real-world human behaviour frequently relies on cognitive heuristics, intuitive judgments, and contextual adaptations (Kahneman, 2011; Tversky & Kahneman, 1974). This reflective essay critically evaluates two personal experiences: a major life decision involving cross-border relocation and a complex workplace problem-solving scenario concerning digital tool deployment. By comparing initial naïve approaches with module-informed theoretical strategies, this paper critically assesses how cognitive frameworks, including System 1/System 2 processing, bounded rationality, and problem restructuring, can enhance strategic outcomes while acknowledging their real-world operational limitations (Simon, 1970; Sternberg, 2003).

Part 1: Decision-Making Scenario – Relocating to the UK

Section 1: Naïve Approach

In 2019, social and political shifts in Hong Kong created significant turbulence, culminating in the introduction of a new UK immigration pathway in 2020. Having harboured a long-standing desire to live abroad, compounded by growing anxiety over the local political climate, limited long-term career prospects, and dissatisfaction with toxic interpersonal dynamics in the local work culture, I made the swift decision to migrate to the United Kingdom.

At the time, my decision-making process was largely unstructured. I did not systematically compare alternative destination countries, nor did I undertake an exploratory visit to the UK. Instead, I relied primarily on my long-standing desire to leave Hong Kong and the fact that an acquaintance had already settled in my eventual destination city. I believed that a short visit would provide only a limited impression of everyday life and would not necessarily predict my long-term experience. I was also concerned that spending too much time evaluating alternatives might weaken my determination to proceed with such a significant life change. Consequently, I selected my destination based largely on familiarity and personal confidence that I would adapt after relocation, despite having conducted little formal comparison between different regions of the UK.

After arriving in the UK, I found that another migrant living nearby had previously undertaken much more extensive research and had also studied in the UK before relocating permanently. Nevertheless, we eventually settled in the same neighbourhood. At the time, the only aspect of my preparation that I recognised as inadequate was the size of my financial reserve, which would have provided greater security during the transition period.

Section 2: Module-Informed Approach

Reconsidering this decision after completing this module, I recognise that my approach was driven primarily by intuition and personal conviction rather than a structured evaluation of alternatives. Looking back, this reliance on instinct resembles Gigerenzer's (2007) concept of gut feelings under conditions of uncertainty. My judgement was largely relied heavily on System 1 thinking which is an intuitive, fast, and emotionally driven cognitive process (Halpern & Dunn, 2023; Kahneman, 2011). The heightened emotional impact of the socio-political situation triggered a strong framing effect, leading me to focus mainly on the imperative of leaving Hong Kong while underestimating a comprehensive, evidence-based evaluation of the destination (Halpern & Dunn, 2023).

Had I applied Halpern and Dunn's (2023) four-step critical thinking framework, I would have engaged System 2 analytical processing to systematically structure the decision. In the goal-setting phase, instead of defining the objective simply as "relocation," I would have established a multi-attribute goal matrix encompassing physical security, psychological well-being, long-term career sustainability, and financial adaptability. To reduce confirmation bias (Nickerson, 1998) and the availability heuristic (Tversky & Kahneman, 1974), I would have conducted a systematic data-gathering process regarding local labour markets, cost-of-living indices, and public infrastructure across multiple UK regions (Halpern & Dunn, 2023). Furthermore, during option evaluation, I would have employed Tversky's (1972) Elimination by Aspects model, setting strict criteria such as professional skill alignment and capital burn rates to systematically rule out unsuitable regions (Halpern & Dunn, 2023). Finally, to address risk management, I would have constructed a worst-case financial scenario model, calculating exact reserve depletion rates under potential unemployment to ensure financial resilience.

Section 3: Differences and Critical Reflection

The module-informed framework introduces clear advantages over the naïve approach, particularly regarding financial forecasting and risk mitigation. By formalising goals and conducting structured trade-off analyses, the systematic approach reduces cognitive uncertainty and emotional distress during the transitional phase.

However, a critical evaluation reveals that relying on intuitive decision-making under high-stakes conditions is not entirely flawed. As Simon (1969, 1970) highlighted in the theory of Bounded Rationality, human decision-makers operate under severe information and time constraints, making optimal rationality unattainable in complex environments. The unpredictable socio-economic changes observed in the UK over recent years demonstrate that even the most rigorous pre-migration models cannot fully account for external volatile shifts. Over-relying on System 2 deliberation risks inducing analysis paralysis, potentially stalling decisive action when time-sensitive opportunities arise (Halpern & Dunn, 2023). Thus, while theoretical frameworks significantly enhance procedural execution and risk defence, person-

al intuition and value-driven choices remain indispensable when navigating fundamental life transitions.

Part 2: Problem-Solving Scenario – Digital Tool Deployment Obstacles

Section 1: Naïve Approach

In a professional context, I designed and developed a digital application aimed at streamlining operational workflows within my local site. Without adequate cross-departmental alignment, the internal Continuous Improvement (CI) team abruptly decided to mandate the rollout of this tool across all national sites. As a junior, non-local female employee, I encountered pronounced organisational resistance during implementation. Several senior managers at other sites, mainly white, male, and holding implicit cultural authority, demonstrated strong skepticism, while the CI manager leading the initiative lacked basic process knowledge and engaged in defensive responsibility-shifting when cross-site conflicts emerged.

My initial response relied on a defensive boundary-setting strategy. Recognising my lack of formal authority and feeling powerless to challenge the entrenched power dynamics and special organisational culture, I restricted my contribution strictly to technical support. I explicitly informed both the CI manager and site leads that my scope was limited solely to resolving software bugs, delegating all challenges related to change management, stakeholder communication, and governance back to the CI lead. While this protective boundary shielded me from personal burnout, it ultimately resulted in a semi-stagnant deployment process, leaving systemic organisational friction unaddressed.

Section 2: Module-Informed Approach

Analysing this scenario through cognitive problem-solving theories reveals that my initial approach reflected a narrow definition of the "Problem Space" via Means-Ends Analysis (Simon & Newell, 1970). I focused almost exclusively on closing technical performance gaps, failing to recognise that the core issue was fundamentally an organisational and human dynamics problem rather than a technical one (Halpern & Dunn, 2023; Simon & Newell, 1970).

Applying Halpern and Dunn's (2023) framework on problem representation and root cause analysis, a module-informed approach requires restructuring the problem space from "tool deployment" to "stakeholder management and change governance." To address systemic resistance without internalising blame, I would implement proactive structural safeguards prior to execution by incorporating structured change management principles (Kotter, 1996) and problem-solving dynamics (Sternberg, 2003). First, I would insist on a formal project charter that explicitly establishes the CI team's governance accountability, thereby securing official organisational authorisation to legitimise my role. Second, I would formulate a Standard Operating Procedure (SOP) paired with a phased pilot approach, decomposing the overarching goal into manageable sub-goals (Halpern & Dunn, 2023; Simon & Newell,

1970). By conducting stakeholder mapping and establishing clear SOPs early on, this strategy replaces passive technical troubleshooting with structured preventative communication, ensuring that operational boundaries and expectations are transparently defined.

Section 3: Differences and Critical Reflection

Comparing the two approaches highlights the clear efficacy of the module-informed framework. By re-framing the problem space, securing formal authorisation, and deploying SOPs, I can transition from passive technical defence to proactive process leadership, preventing organisational dysfunctions from translating into personal psychological burdens.

Critically reflecting on this transformation reveals substantial transferable value for my future career development. Particularly in prospective customer-facing roles, where guiding external clients and altering established operational habits involve heightened resistance, leveraging stakeholder management and SOPs serves as a crucial mechanism to define realistic boundaries early and prevent misaligned expectations. Nevertheless, theoretical frameworks possess inherent limitations. Simon's (1969) model assumes problem-solving occurs within rational organisational systems. In reality, workplace environments are frequently skewed by implicit biases, structural inequalities, and political manoeuvring (Pfeffer, 1992). These real-world constraints mean that even a well-structured problem-solving framework cannot single-handedly overcome distorted organisational cultures without sustained structural support from executive leadership.

Conclusion

In conclusion, critically reflecting upon personal decisions and problem-solving scenarios reveals the profound utility of cognitive theoretical frameworks. Integrating System 2 deliberative structures, decision matrices, and strategic problem-space re-representations enables individuals to transcend emotional heuristics, mitigate systemic biases, and establish clear operational boundaries (Halpern & Dunn, 2023; Simon & Newell, 1970). However, normative cognitive models must be tempered with real-world pragmatism; bounded rationality, temporal volatility, and structural organisational power dynamics often constrain pure analytical optimisation (Pfeffer, 1992; Simon, 1969). This module has not changed my values, but it has changed how I structure important decisions and complex workplace problems. Rather than relying solely on intuition, I now recognise the value of deliberately slowing down, defining goals clearly, and evaluating alternatives systematically before acting.

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